# Cloud Storage Troubleshooting & Operational Guide

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## 1. S3 Bucket 403 AccessDenied Error Troubleshooting

HTTP 403 AccessDenied errors typically stem from misaligned IAM policies, S3 Bucket Policies, or KMS access restrictions. To resolve: (a) Check user/role IAM permissions for 's3:GetObject', 's3:PutObject', and 's3:ListBucket'. (b) Verify S3 Bucket Policy does not contain explicit Deny statements that override role permissions. (c) Ensure VPC Endpoint policies allow traffic from the caller's VPC. (d) For cross-account access, verify that both the source IAM role and target bucket policy explicitly permit the action.

## 2. IAM Permissions & Bucket-Policy Verification

Run the policy simulator command 'aws iam simulate-principal-policy' to test action evaluation against the target ARN. Verify that S3 Public Access Block settings are enabled on buckets holding private data, and confirm AWS Organizations SCPs do not restrict S3 actions in the designated AWS region.

## 3. AWS KMS Key Rotation Guidance

Enterprise storage buckets are encrypted server-side using customer-managed AWS KMS keys (SSE-KMS). Customer-managed KMS keys rotate automatically every 365 days (1 year) when automatic key rotation is enabled. During rotation, previous key material is retained to decrypt existing ciphertexts. Ensure calling principals possess 'kms:Decrypt' and 'kms:GenerateDataKey' permissions on the KMS key policy.

## 4. Storage Upload Limits and Object Sizing

Maximum single-upload object size: The maximum allowable file size for a single S3 PUT upload operation is strictly 50 GB. For objects exceeding 50 GB and up to the total storage ceiling of 5 TB, clients must use the AWS S3 Multipart Upload API. Multipart uploads recommend part sizes between 10 MB and 100 MB for optimal network recovery and transfer velocity.